

Triangle inequality

Proposition (Triangle inequality)

$$u, v \in \mathbb{R}^n$$

$$\|u + v\| \leq \|u\| + \|v\|$$

This is a consequence of the Cauchy-Schwarz inequality.

Pf:

$$\begin{aligned} \frac{\|u+v\|^2}{1} &= (u+v) \cdot (u+v) = u \cdot u + v \cdot u + u \cdot v + v \cdot v \\ (\text{LHS})^2 &= \|u\|^2 + 2u \cdot v + \|v\|^2 \end{aligned}$$

← key comparison

$$(\text{RHS})^2 = (\|u\| + \|v\|)^2 = \|u\|^2 + 2\|u\| \cdot \|v\| + \|v\|^2$$

$$2u \cdot v \leq 2|u \cdot v| \leq 2\|u\| \cdot \|v\|$$

Cauchy-Schwarz

Generalized eigenspaces

Defⁿ $A \in M_{n \times n}(F)$, $\lambda \in F$

Generalized λ -eigenspace

for A is the subspace

$$E_{\lambda}^{\text{gen}} = \left\{ v \in F^n : (A - \lambda I_n)^m \cdot v = 0 \right. \\ \left. \text{for some } m \geq 1 \right\}$$

Prop: $v_1, v_2 \in E_{\lambda}^{\text{gen}}$

$$\left. \begin{array}{l} (A - \lambda I_n)^{m_1} \cdot v_1 = 0 \\ (A - \lambda I_n)^{m_2} \cdot v_2 = 0 \end{array} \right\} \Rightarrow (A - \lambda I_n)^{\max\{m_1, m_2\}} \cdot (v_1 + v_2) = 0$$

Theorem $F = \mathbb{C}$

There is a basis for $\mathbb{C}^n$ consisting of
generalized eigenvectors for A

Example

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$(A - 2I_3) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} : \begin{array}{l} e_1 \mapsto 0 \\ e_2 \mapsto e_1 \\ e_3 \mapsto e_2 \end{array}$$

$$(A - 2I_3)^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} e_1 \mapsto 0 \\ e_2 \mapsto 0 \\ e_3 \mapsto e_1 \end{array}$$

$$(A - 2I_3)^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Translation of the theorem

There is a basis B for $\mathbb{C}^n$ s.t. the B -matrix for $v \mapsto A \cdot v$ is given by a matrix of

the form

$$\begin{bmatrix} \lambda_1 & 1 & 0 & & & & & & \\ & \lambda_1 & 1 & & & & & & \\ & & \lambda_1 & 1 & & & & & \\ & & & \lambda_2 & 1 & 0 & & & \\ & & & & \lambda_2 & 1 & 0 & & \\ & & & & & \ddots & \ddots & \ddots & \\ & & & & & & \lambda_r & 1 & \\ & & & & & & & & \lambda_r \end{bmatrix}$$

This is a matrix in Jordan canonical form

Review

- Systems of linear equations
 - Sets, functions, induction.
- Fields ($\mathbb{Z}/p\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$)
- Vector spaces (F^n , $\text{Poly}(F)$, $M_{n \times m}(F)$)
- Linear transformations ($T: V \rightarrow W$)

Final abstraction:

$$T: V \rightarrow W \ni w$$

Solve: $T(x) = w$

- Vector spaces:
 - Subspaces ($\ker T$, $\text{im } T$)
 - Linear independence
 - Span.
 - Basis
 - Dimension (Exchange Lemma)

• Linear transformations & matrices: $T: V \rightarrow W$

• Rank-nullity $\left(\begin{array}{l} \dim \ker T + \dim \operatorname{im} T \\ = V \end{array} \right)$

• $T: V \rightarrow V \leadsto [T]_{\mathcal{B}} \in M_{n \times n}(F)$
 $\mathcal{B}$: basis for V
 $n = \dim V$

• T is an isomorphism $\leadsto [T]_{\mathcal{B}}$ is invertible

• $T(v) = w \leadsto [T]_{\mathcal{B}} y = w$

This also holds for $T: V \rightarrow W$
bases $\mathcal{B}_1$ for V
 $\mathcal{B}_2$ for W

$\leadsto [T]_{\mathcal{B}_1, \mathcal{B}_2} y = w$